

Proposed Solution and Architecture

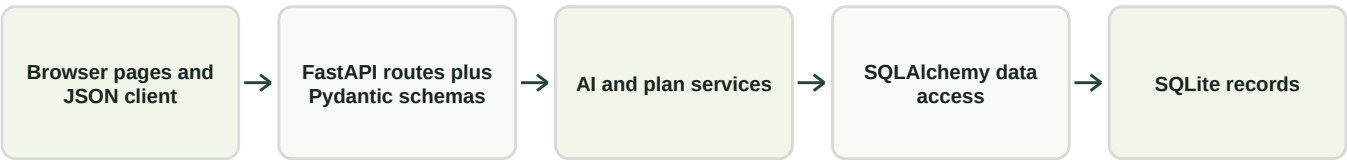
FitBuddy AI Fitness Plan Generator

A component-level view of the proposed FitBuddy solution.

Proposed solution

FitBuddy is a local-first FastAPI application with a responsive browser interface and JSON API. It validates a profile, generates a weekly plan and general tip through Gemini or the configured demo path, stores the result in SQLite, and supports feedback-based revision.

Architecture overview



The result is rendered as a web page or returned as JSON.

Component responsibilities

Component	Responsibility
app/main.py	Create the FastAPI application, mount static assets, register routes and initialize database tables.
app/routes.py	Handle the browser form, feedback, coach pages and API endpoints.
app/schemas.py	Define input rules for profiles and feedback and the accepted data shapes.
app/services/ai.py	Call Gemini or provide deterministic demo responses, including safe fitness guidance.
app/models.py and app/database.py	Define SQLite records, relationships, engine and sessions.
app/templates and app/static	Render the profile, plan and coach pages and responsive styles.

Safety and trust controls

- Treat profile and feedback text as user data, not instructions that can override safety rules.
- Keep the app local for the prototype and use private credentials.
- Do not expose admin tokens in URLs or logs in any hosted environment.
- Show general-information boundaries and direct health concerns to qualified professionals.